

Hyperledger Fabric: Tailored to Scale Blockchain Applications on IBM Cloud

This article discusses the blockchain and Hyperledger Fabric, and how the two can be used to build and operate an efficient software solution on IBM Cloud.

According to the blockchain statistics released by FinancesOnline, by February 2021, around 620.37 million blockchain transactions had been recorded globally, with North America accounting for 46 per cent of these. Banking topped the list of global top blockchain investors with a 29.7 per cent share, followed by process manufacturing and discrete manufacturing with 11.4 and 10.9 per cent, respectively. According to current market trends, the United States and Western Europe will invest around US\$ 4.2 billion and 2.9 billion in blockchain technologies, respectively, by the year 2022.

As per the 2020 Blockchain Global Patent Authorization report, 3924 blockchain technology patents have been filed, with the US leading at 39 per cent, Korea at 21 per cent and China at 19 per cent. Alibaba leads with 212 blockchain patents, followed by IBM's 136.

IBM has been an initiator in so many ways – hiring the first disabled employee way back in 1914, building the IBM baseball team in 1929, hiring women systems service professionals in 1935, developing the commercial electronic multiplier in 1946, setting up its first office in India in 1951, and so on. Inspired by the success of blockchain and just a decade and a half into the 21st century, IBM has contributed to Hyperledger Fabric.

Blockchain and Hyperledger

As we work our way through the maze of technology, we'll start with the basics: blockchain and ledger. Storage of data was previously restricted mostly to traditional databases. Blockchain evolved as an information recording system. The data in a blockchain is updated and synchronised in real time. Its inherent characteristic is its robustness against hacking and fraud.

The blockchain is neither supervised nor managed by a person, thus making it a decentralised entity. Everyone can access it; thereby, it is permissionless. Every transaction is a timestamped record, and complete proprietorship of records is maintained in a fully confidential manner. A novel way of writing data into blockchain involves encoding the data as an address and later directing a transaction to that receiver address instead of using the payload. Advanced cryptographic algorithms ensure complete safety of the data inside the blockchain. It is due to this encryption and verification that blockchains are comparatively slower than databases. Blockchain is lauded in the financial and business worlds, where decentralisation, monetary transactions, and trusted data verification are top priorities. It provides a platform to verify and audit transactions through multiple supply chain partners. Every new day brings advancements in blockchain technology (BCT).

As per Deloitte's Global Blockchain Survey (2020), the penetration of blockchain technologies in production has jumped from 23 per cent in 2019 to 39 per cent in 2020. Sixty-four per cent of the respondents opted for enterprise controlled digital assets, 59 per cent for decentralised cryptocurrencies, and 54 per cent for algorithm driven stablecoins. According to the same survey, respondents from China, Brazil

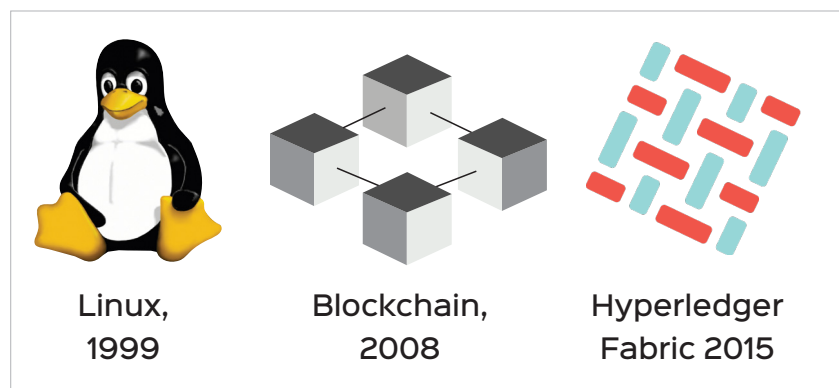


Figure 1: The evolution of Hyperledger Fabric